

Project Insight — *Expectation Decider*

Purpose of the Analysis

This project analyzes how **study hours**, **attendance**, and **group discussion participation** influence a student’s probability of **passing the final exam**. Using real data from 200 students, we applied probability and statistics concepts to uncover meaningful academic patterns.

What the Data Reveals

- Many students (**158**) study more than 10 hours per week.
- Only **72** students maintain attendance above 80%.
- Just **83 out of 200** students passed the final exam (overall pass rate **41.5%**).
- **99** students participate in group discussions.

These numbers already hint that **effort alone (study hours)** may not be the only factor behind success.

Insight #1 — Group Discussion Has the Strongest Impact

Category	Pass Rate
Overall students	41.5%
Students in group discussion	66.7%

Students who engage in group discussions are **much more likely to pass** than the average student.

Interpretation:

Active participation, peer learning, and discussion improve understanding more than isolated studying.

Insight #2 — More Study Hours ≠ Guaranteed Success

Even though the majority study more than 10 hours weekly, the pass rate remains low.

Interpretation:

Studying longer is not as effective as **studying smarter** (interactive learning).

Insight #3 — Attendance Improves Probability

Using Bayes’ theorem (given values), we found:

$$[\\ P(\text{Pass} \mid \text{High Attendance}) = 80.3\% \\]$$

Students with high attendance have a **very high chance of passing**.

Interpretation:

Regular presence in lectures supports performance, but engagement still matters more.

Insight #4 — Events Are Dependent (Real-Life Proof)

Since:

$$[\\ P(\text{Pass} \mid \text{Discussion}) \neq P(\text{Pass}) \\]$$

Passing the exam **depends** on whether a student participates in discussions.
This shows practical application of **dependent events** from probability theory.

Insight #5 — Probability Distribution of Random Selection

When selecting 3 random students:

- Most likely outcome is **1 or 2 students passing**
- Very low chance all 3 pass

Interpretation:

This reflects the overall difficulty level of the exam and the low pass rate.

Final Conclusion

This project clearly demonstrates:

Interactive learning (group discussion) + good attendance influence exam success more than simply spending long hours studying.

Most Influential Factor: Group Discussion Participation